Pre-analysis plan: #ManyDaughters Impact of Having Daughters on Attitudes, Preferences, and Behaviors

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This is a pre-analysis plan for testing the four hypotheses of the #ManyDaughters project based on the data from the German Socio-economic panel. We describe the methodology and the variables to be employed for the final analysis. Further on, we deliver the interpretation of potential results and discuss the limitations of the analysis.

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1 H1

Hypothesis 1: Having daughters leads to stronger support for gender equality (as measured by attitudes). That is, denoting X as a measure of support for gender equality,

(X for respondents with daughters) > (X for respondents without daughters)

1.1 Specification of the regression model

In the ideal scenario where attitudes towards gender norms are surveyed annually in a panel context, we would suggest analyzing individual within variation with a fixed effects (FE) model to consistently measure the average treatment effect (ATT) on attitudes after becoming a parent to a daughter. However, in this section, we cannot utilize SOEP's longitudinal nature, as gender attitude variables are only available for the survey year 2018. Thus, we restrict the methodology to estimating a linear model with OLS based on cross-sectional data. In the following, we will explain why this might still lead to causal effects of having a daughter on gender attitudes. In this context, respondents with at least one daughter constitute our treatment group. As a control group, we choose parents who do not have any daughters. Thus, the final sample consists of only parents.

To test H1, we estimate the following linear model with OLS:

$$X_i = \beta_0 + \beta_1 d_i + A_i'\delta + \epsilon_i \tag{1}$$

where X_i is the outcome variable for the attitude regarding gender equality (GE index) for individual i, d_i indicates whether individual i has at least one daughter, ϵ_i is a random error. A_i includes additional demographic characteristics (see **Table 1**).

We calculate the GE index based on responses to three items following Hamjediers (2021). These three items are described in section 1.3.1:

1. *Children below the age of 6 suffer if their mother works*.
2. *A child under the age of three suffers if his or her mother works*.
3. *It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally* (reverse-coded).

For the GE index, we add all valid survey responses to each of the three above items with equal weights and min-max-normalise the index.

Table 1: Control Variables H1

Variable	Description
age_i	Age of the respondent i in years. Literature finds more traditional gender role attitudes among older participants (see, e.g., Lynott and McCandless, 2000).
mig_i	$= 1$ if respondent i has migration background; $= 0$ otherwise. Migrant adolescents in Germany tend to have more traditional gender role attitudes than their native peers Kretschmer (2018).
voc_i and uni_i	$= 1$ if respondent i has vocational training; $= 0$ otherwise. respectively, $= 1$ if respondent i has a university degree; $= 0$ otherwise. "no vocational training" is the reference category. Du et al. (2021) find that the extra schooling leads to more egalitarian gender role attitudes. Rivera-Garrido (2022) finds that for women from a poor family background, education reduces the probability of agreeing with traditional gender norms by more than 11 percentage points. We do not include income as control variable because of high correlation with education.
$east_i$	$= 1$ if respondent i lives in East Germany; $= 0$ otherwise. The literature suggests that there are substantial East-West differences in attitudes toward the traditional gender division of labor among German women and men. East Germans are more likely than West Germans to oppose a gender-based division of labor (Adl, 1996). East Germans are significantly more likely to hold egalitarian sex-role attitudes than West Germans (Bauernschuster and Rainer, 2012). Hamjediers (2021) also finds differences in gender ideologies between East and West Germany, with regions of the former German Democratic Republic (GDR) being less traditional.
$female_i$	$= 1$ if respondent i is female; $= 0$ if male. According to Horne and Johnson (2018), females score lower than males on traditional gender role attitudes. Aassve et al. (2014) also find that men have on average more traditional gender role attitudes than women. In contrast, Düval (2023) finds no gender difference in attitudes.
$religiosity_i$	$= 1$ if respondent i is religious; $= 0$ otherwise. Morgan (1987) finds that religious devoutness is the most important variable predicting traditional gender-role attitudes. As religious devoutness increases, gender-role attitudes become more traditional.
$married_i$ $separated/div_i$ $widowed/single_i$	Dummy variables capturing marital status. Respectively $= 1$ if respondent i is married, separated/divorced or widowed/single; $= 0$ otherwise. Married individuals tend on average to display more traditional views (Brewster and Padavic, 2000). See also, e.g., Perales et al. (2018).
$polit_i$	Political ideology normalized to scale $[0, 1]$. See also section 2. Oswald and Powdthavee (2010) finds that first time fathers of daughters move toward left-wing political parties, and that having a son leads to favoring more right-wing parties.

1.1.1 Potential concerns regarding the causality

Concerns regarding the causality in our approach may arise mainly if the variation in X_i is not exogenous. In this case, the OLS estimator may produce biased estimates. We will discuss some potential threats to our identification. Firstly, we are less concerned about exogeneity of d_i as having a daughter is an exogenous event (in contrast to not having a child (!)), i.e., the assignment of child sex is nearly random (Takaku, 2018). Because of this, reversed causality should also not be a concern. Thirdly, the omitted variable bias would arise if the error term is partially correlated with a variable in the model and is simultaneously correlated with the dependent variable. We hope that this is less likely in case of d_i. Furthermore, we account for additional control variables. Lastly, the selection problem could arise if we had responses for attitudes only for parents. This is not the case, as the gender equality attitudes have been inquired from all respondents regardless of their parenthood status.

1.2 Interpretation of results

The main parameter of interest is the coefficient β_1 in equation 1, which captures the average difference in gender equality attitudes between parents with and without at least one daughter, conditional on covariates. Let $\hat{\beta}_1$ denote the estimated coefficient.

If $\hat{\beta}_1$ is not statistically significant (at conventional levels: * $p < 0.10$, ** $p < 0.05$, ***$p < 0.01$) or is statistically significant but less than or equal to zero, the data do not support Hypothesis H1, and the hypothesis should be rejected. Conversely, if $\hat{\beta}_1 > 0$ and statistically significant, the data do support H1, and we would accept the hypothesis accordingly.

1.3 SOEP Variables to be used

The SOEP contains multiple codes indicating missing or invalid responses. These include, but are not limited to, the following: -1 for "no answer", -2 for "does not apply", -3 for "answer improbable", -4 for "inadmissible multiple response", -5 for "not included in this version of the questionnaire", -6 for "version of questionnaire with modified filtering", -7 for "only available in less restricted edition", and -8 for "question not part of the survey program in the given year". In the process of operationalizing constructs, these values are generally treated as missing.

Unless otherwise specified, variables are taken from the survey year 2018, which corresponds to the year in which our dependent variable was measured.

This ensures temporal consistency in the operationalization of constructs and alignment with the outcome data.

We identify parents based on the BIOBIRTH, the biological Parent/Child Connections module of the SOEP (Zimmermann and Saalbach, 2024). It contains fertility information on every woman and man who has ever provided at least one successful SOEP Biography interview. It provides person IDs, year of birth, and sex for up to 19 children of each person.

1.3.1 Dependent Variable

The dependent variable *Support for Gender Equality* is constructed based on a set of survey items included in the 2018 questionnaire. This item battery captures respondents' attitudes toward gender roles and equality through their level of agreement on a seven-point Likert scale, ranging from 0 (strongly disagree) to 7 (strongly agree). The included items focus on perceptions related to parental employment and its impact on young children, as well as beliefs about the division of labor between men and women in the household.
Specifically, the items used to operationalize this variable are:

- Children below the age of 6 suffer if their mother works. [bip_197_04]
- A child under the age of three suffers if his or her mother works. [bip_197_05]
- It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally (*reverse-coded*). [bip_197_06]

An additive index is constructed by summing the responses to the three selected items. Since the third item is phrased in the opposite direction, it is reverse-coded before summation to ensure consistency in the index. Each item is measured on a 7-point Likert scale ranging from 0 to 7, resulting in a raw index score ranging from 0 to 21. To facilitate interpretation and comparability, this raw score is subsequently normalized to a scale between 0 and 1 using min-max normalization:

$$GE\ index = \frac{Additive\ Index - Min}{Max - Min},$$

where *Min* and *Max* represent the minimum and maximum possible values of the additive index, respectively. The resulting normalized index thus ranges from 0 (indicating the lowest possible support for gender equality) to 1 (indicating the highest possible support).
In addition to the three items used to operationalize "Support for Gender Equality", the item battery also included several other items related to family and relationship norms. These items encompass attitudes towards marriage among

long-term partners, legal recognition of same-sex marriages, parenting capabilities of single parents and same-sex couples, as well as societal acceptance of transgender individuals. Although these items provide valuable insights into attitudes toward sexual orientation, family structures, and transgender rights, they were deliberately excluded from the construction of the Support for Gender Equality index. This decision was made because these items focus more specifically on issues of sexual orientation and LGBTQ+ rights rather than on the broader concept of gender equality related to the division of labor and gender roles within the family and society. Including them could therefore confound the measurement of attitudes explicitly centered on gender equality.

1.3.2 Independent Variables

- *daughter*
  The definition of having a daughter is based on the BIOBIRTH module of the SOEP (see Zimmermann and Saalbach, 2024). It identifies up to nineteen biological children of every person in the SOEP and their sex. If one of the children is a daughter, we code the daughter indicator $= 1$, and $= 0$ otherwise. We consider all individuals who had a daughter by the time of the 2018 interview. In doing so, we take into account the birth year, birth month, and sex of the children.

- *age*
  The age of respondents is derived from variable *gebjahr* (year of birth). *age* is calculated as the difference between the year of our dependent variable 2018 and *gebjahr*. If *gebjahr* takes the value -1 (indicating a missing or invalid response), the derived variable *age* is coded as missing accordingly.[1]

- *mig*
  The variable *migback* (Migration Background) is recoded into a binary variable *mig*. Respondents without a migration background (coded as 1 in *migback*) are assigned a value of 0 in *mig*. Respondents with a direct (2), indirect (3), or undifferentiated (4) migration background are assigned a value of 1 in *mig*.

- *voc*
  The variable *pgpbbil01* (Vocational Degree Received) is recoded into a bi-

[1] For a robustness check, the continuous age variable is recoded into a categorical variable. The following age groups are defined: (1) Missing (age not available), (2) 18–39 years, (3) 40–59 years, and (4) 60 years and older. This reduced classification allows for non-linearity and captures broad life stages while ensuring sufficient group sizes and allows testing the robustness of results across age groups.

nary variable *voc*. Respondents who report having completed a vocational degree (codes 1 through 7) are assigned a value of 1 in *voc*. Respondents without a vocational degree (code -2: "does not apply") are assigned a value of 0.

- *uni*
  The variable *pgpbbil02* (College Degree) is recoded into a binary variable *uni*. Respondents who report having obtained a college or university degree (codes 1 through 10) are assigned a value of 1 in *uni*. Respondents without such a degree (code -2: "does not apply") are assigned a value of 0.

- *east*
  The variable *east* is derived from the respondent's current region of residence, as recorded in the variable *sampreg*, which distinguishes between West Germany (coded as 1) and East Germany (coded as 2), based on the borders as of 1990. For respondents living in Berlin, East-West classification is approximated using postal codes. In the recoded binary variable *east*, respondents residing in East Germany are assigned a value of 1, while those residing in West Germany are assigned a value of 0.

- *female*
  We determine the gender of respondents using variable *sex*, which records the sex assigned to individuals at birth. If a respondent indicates a value of "1 = male" for the variable *sex*, the derived variable *female* is coded as 0. If the respondent indicates a value of "2 = female", *female* is coded as 1. Respondents with responses such as "no answer", "does not apply", or missing values on *sex* are assigned a missing value on *female*.

- *religiosity*
  Religiosity is measured using the item "Religion Important for Well-Being", which was surveyed in 2017. Respondents were asked to indicate how important religion is for their well-being, with the following response options: (1) Very important, (2) important, (3) less important, (4) unimportant. For the analysis, the variable is recoded into a binary indicator: respondents who selected "very important" or "important" are coded as 1, while those who selected "less important" or "unimportant" are coded as 0. We assume a longitudinal persistence in religious beliefs and transfer the values from 2017 to our main analysis sample of 2018 for the available individuals.

- *married*
  Marital status is measured using the variable *pgfamstd* ("Marital Status in

Survey Year"), which captures the institutional status of marriage at the time of the individual interview. It is based on information provided by the respondent about their current relationship status, complemented by retrospective data from the biography questionnaire.
The variable *married* is coded as 1 if the respondent is either married (category 1) or in a registered same-sex partnership and living together (category 7), and 0 otherwise.

- *separated/divorced*
  The variable *separated* is also based on *pgfamstd* and is coded as 1 for individuals who are married but separated (category 2) or in a registered same-sex partnership but living apart (category 8), or divorced (category 4) and 0 otherwise.

- *widowed/single*
  The variable widowed/single is likewise constructed using *pgfamstd*. It is coded as 1 for individuals who are single (category 3), or widowed (category 5), including those whose same-sex partner has passed away, and 0 otherwise.

- *polit*
  The dependent variable "political attitude" [plh0004] is measured on an 11-point scale ranging from 0 to 10, where 0 represents the extreme left and 10 represents the extreme right. Intermediate values correspond to gradations along the left-right political spectrum. The question was included in the survey waves of 2005, 2009, 2014, and 2019. As the variable is not available in 2018, we will impute it from the individuals in 2018, from years 2014 and 2019 if 2014 values are not available.

1.4 Inclusion/exclusion of observations

In the analysis of this section, we include only respondents with at least one child, i.e., respondents who are parents.Although having a daughter is regarded as an exogenous event, having a child generally is not. Becoming a parent entails a variety of selection processes.[2] Thus, comparing parents and non-parents might be inappropriate. For the discussion of endogenous preferences on fertility see also Hashimzade (2020) or Diaz et al. (2011). Furthermore, we allow the possibility of same-sex households in the data.

[2]However, we acknowledge that becoming a parent can still be an exogenous event in minority of cases.

The analysis is restricted to the year 2018 and to respondents for whom the responses to the three gender attitude items from section 1.1 are available. For the robustness check regarding the selection due to missing values in the control variables, we will estimate a separate model with indicators for missing values for the respective variables, and compare the results with the baseline regression equation 1. We also exclude all respondents with incomplete data on the gender roles.

2 H2

Hypothesis 3: Having daughters leads to more left-leaning political preferences. That is, denoting Y as a measure of left-leaning political preferences,

$$(Y \text{ for respondents with daughters}) > (Y \text{ for respondents without daughters})$$

2.1 Specification of the regression model

For the given section, we exploit the panel structure of the SOEP (for the years 2005, 2009, 2014, 2019) and analyze individual within variation in political preferences after becoming a parent to a daughter. We estimate the following linear Fixed effects model:

$$Y_{it} = \gamma_0 + \gamma_1 d_{it} + B'_{it}\eta + year_t + \alpha_i + \nu_{it} \tag{2}$$

where, the outcome variable Y_{it} is a measure of left-leaning political preferences for individual i in year t. It is measured on a eleven-point Likert scale, which we normalize. d_{it} indicates whether individual i has at least one daughter in year t, α_i is individual fixed effect, ν_{it} is a random error term including the unobserved heterogeneity. Matrix B_{it} includes other individual-level control variables found in **Table 2**. If the data is not sufficient regarding the changes in d_{it}, then we will estimate a pooled OLS regression with year fixed effects.

One drawback of the FE estimation is that the explanatory variable cannot be time-constant for all observations. If we have a panel of adults and one explanatory variable is education, we can allow education to be constant for some part of the sample. But we must observe education changing for some people in the sample (Wooldridge, 2010).

Respondents with at least one daughter constitute our treatment group. As a control group, we include parents who do not have any daughters. Thus, the final sample consists of only parents, i.e., only individuals who at some point become parents of a daughter. The FE estimator uses only within variation. Thus, the Fixed Effects (FE) estimator can only be applied when there is variation within the units being studied. Since the FE estimator relies solely on within-unit variation, its generalization is limited to those units that exhibit such variation (those who, at some point, become parents). For these observations, the FE estimator provides an average treatment effect for the treated (ATT).

Table 2: Control variables H2

Variable	Description
age_i	Age of the respondent i in years. Younger voters tend to prefer more left-wing and progressive parties, whereas older voters tend to favor parties that are more conservative (Stockemer and Sundström, 2025). Consult also, e.g., Fisher (2008) and Carroll et al. (2018) on the age gap in US politics.
mig_i	$= 1$ if respondent i has migration background; $= 0$ if otherwise. Kende et al. (2024) survey political preferences of people living in Germany with a migrant background. Here, the preference formation is influenced by the migrant status (first- or second-generation), dominant values within the sending society, or religious affiliation.
voc_i and uni_i	$= 1$ if respondent i has vocational training; $= 0$ otherwise. Education level and field of study can significantly influence political attitudes. For instance, academic disciplines can stimulate the development or change of political attitudes among students (Junior et al., 2020).
$east_i$	$= 1$ if respondent i has East German background; $= 0$ otherwise. Political preferences can be influenced by the political system individuals experience (e.g. Fuchs-Schündeln and Schündeln, 2015).
$female_i$	$= 1$ if respondent i is female; $= 0$ if male. Women in parliament tend to be more leftist than men (see, e.g., Wängnerud, 2000). More recently, Grasso and Shorrocks (2025) find that for generations born from 1946 onward, women are more supportive of government intervention on inequality, redistribution, and service provision than men. Exley et al. (2025) find that women are believed to be more generous and more equality-oriented than men. See also Gidengil et al. (2003) for the discussion of gender differences in policy preferences.

To identify the causal effect, the approach needs only the assumption that the within variation is exogenous (see section 1.1.1). Our treatment indicator d_{it} is equal to 1 if individual i has at least one daughter in year t, disregarding whether the individual has any other children. In the FE model, the coefficient estimate of d_{it} will not be estimated for individuals without a change in the treatment indicator.

Additionally, to control for constant explanatory variables, we also employ the

Pooled OLS estimator for the linear model:

$$Z_i = \delta_0 + \delta_1 d_{it} + B'_{it}\eta + year_t + \upsilon_{it} \tag{3}$$

where, $\hat{\delta_1}$ would be the in-between estimator of the daughter effect. Here as well, we would control for year fixed effects $(year_t)$.

Alternatively, the random effects (RE) estimator can be applied. The main assumption of the RE method is that the individual effects are not correlated with any of the independent variables. In this case, we will conduct the respective Hausman test (Hausman, 1978) to evaluate the potential endogeneity of individual effects and appropriateness of RE estimation. Lastly, if there is no within-variation in the variable of interest, the FE estimator cannot estimate the respective coefficients. Therefore, we will also potentially use extensions of the RE estimator in our validation analysis, such as the Hausman-Taylor (HT) Estimator (Hausman and Taylor, 1981).

2.2 Interpretation of results

Here, the main coefficient of interest is γ_1. If its estimate $(\hat{\gamma_1})$ is negative and statistically significant (* $p < 0.10$, ** $p < 0.05$, ***$p < 0.01$), then this will speak against H2. We would also reject H2 if the estimate of γ_1 is statistically insignificant. On the contrary, if the estimate is positive and statistically significant, it would support H2.

2.3 SOEP Variables to be used

For the treatment of missing values, see Section 1.3.

2.3.1 Dependent variable

The dependent variable "political attitude" is measured on an 11-point scale ranging from 0 to 10, where 0 represents the extreme left and 10 represents the extreme right. Intermediate values correspond to gradations along the left-right political spectrum. The question was included in the survey waves of 2005, 2009, 2014, and 2019. To facilitate analysis, the raw political attitude scores are min-max normalized to a [0,1] scale.

2.3.2 Independent variables

For the choice of the independent variable, see section 1.

2.4 Inclusion/exclusion of observations

In contrast to section 1, we include responses from multiple years 2005, 2009, 2014, and 2019 in which political preferences were part of the questionnaire. In the FE-Estimation, we include only respondents with at least one change in the daughter indicator over the given years who have responses to the political preferences variable. The analysis sample consists only of individuals who have been treated (ATT).

Additionally, we note that there are other factors that we are not controlling for due to data availability reasons. E.g., distributional preferences have been only part of the questionnaire in the special SOEP Innovation Sample (IS). Individuals' preferences regarding the distribution of resources can shape their political attitudes (see, e.g., Kerschbamer and Müller, 2020). Identity and Policy Preferences. Voter behavior can also be determined by identity preferences (Jenke and Huettel, 2020). Lastly, political ideology is partially heritable, with studies suggesting that about 40% of political ideology can be attributed to genetic factors (Rasmussen et al., 2024).

3 H3

Hypothesis 3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes). That is, denoting Z as a measure of gender equality in the household,

(Z for respondents with daughters) > (Z for respondents without daughters)

3.1 Specification of the regression model

We test H3 on the household level by estimating the following linear FE model:

$$Z_{ht} = \eta_0 + \eta_1 d_h + C'_{ht}\delta + year_t + \lambda_h + \kappa_{ht}, \tag{4}$$

where Z_{ht} denotes the outcome variable measuring gender equality in the household h at year t. This variable is constructed as a difference in absolute hours spent on housework between male and female respondents sharing a household, considering only couples. The use of housework hours as an indicator of gender equality within the household is well established in the literature (see Bianchi, 2000; Bittman et al., 2003; Evertsson and Nermo, 2004; Gupta, 2007; Hook, 2017; Voßemer and Heyne, 2019). For this analysis, we restrict the sample to heterosexual households. The variable d_h is a binary indicator equal to 1 if the household has at least one daughter, and 0 otherwise. λ_h and $year_t$ denote household and year fixed effects, respectively. Lastly, in C_{ht} we include controls for characteristics of both household men and women (see **Table 3**).

Table 3: Control variables H3

Variable	Description
m_age_h, m_age_h	Respective age of the male and female of household h in years. The division of housework changes as couples age and move through life together, specifically after the birth of children (Schulz and Blossfeld, 2006).
m_mig_h, f_mig_h	$= 1$ respectively if respondent i has migration background; $= 0$ if otherwise. Immigrant wives barely adjusted their housework times due to relative or absolute income changes in comparison to natives (Fendel and Kosyakova, 2023).
m_voc_h, m_voc_h, m_uni_h, f_uni_h	$= 1$ respectively if male or female of household h has vocational training or university degree; $= 0$ otherwise. Education is a critical factor in promoting gender equality. Higher education levels are associated with more egalitarian attitudes towards gender roles (Vlassoff, 2007)
$east_h$	$= 1$ if household h is in East Germany; $= 0$ otherwise. Trappe (2016) discusses economic gender equality in East Germany before and after reunification, and compares it between East and West Germany including the division of unpaid work at home.
$f_weekly_hours_h$	Household Woman's weekly work hours. Women's participation in the labor force and their income levels can enhance their decision-making power and contribute to more egalitarian household arrangements (Vlassoff, 2007).

3.2 Interpretation of results

Here, the main coefficient of interest is η_1. If its estimate ($\hat{\eta_1}$) is negative and statistically significant (* $p < 0.10$, ** $p < 0.05$, ***$p < 0.01$), then this would speak against H3. We would also reject H3 if the estimate of η_1 is statistically insignificant. On the contrary, if the estimate is positive and statistically significant, it would support H3.

3.3 SOEP Variables to be used

For the treatment of missing values, see Section 1.3.

3.3.1 Dependent Variable

To measure housework hours, we use self-reported time spent on a typical weekday on housework (washing, cooking, cleaning) [pli0043_h], care for persons in need [pli0046], and running errands (e.g., shopping, administrative tasks) [pli0040]. Following Fendel and Kosyakova (2023), we exclude childcare from

the main analysis but include it in robustness checks, as it differs from housework in terms of bargaining relevance, emotional value, perceived enjoyment, and simultaneity with other tasks.
For each household, we sum the absolute number of reported hours for women and men separately and calculate the difference as:[3]

$$Z_{ht} = \text{hours}_{ht,male} - \text{hours}_{ht,female}.$$

The theoretical range of the resulting variable Z_{ht} extends from -72 to 72, assuming a maximum of 24 reported hours per activity across three categories and per person. In the literature, the difference is usually also called the Housework Gap (see Mandel and Lazarus, 2021)

3.3.2 Independent Variables

The variables age, mig, voc, and uni are calculated separately for the man and the woman, with the corresponding prefix m_ or f_ appended. For the operationalization of these independent variables as well as the household-level variable $east$, see Section 1.
In addition, *f_weekly_hours* is included as an independent variable in the model.

- *f_weekly_hours*
  The variable *f_weekly_hours* is based on the item *plb0186_h* ("Actual Weekly Working Hours"), which captures the respondent's actual number of hours worked per week. This variable was surveyed between 1984 and 2019 and is used to measure the weekly working hours of the woman in the household.

3.4 Inclusion/exclusion of observations

The housework hours variable is available in the SOEP for all available years from 1984 to 2022. We only concentrate on male-female households and gender gaps in their housework hours as the hypothesis deals concerns gender equality.

In this section, we do not specifically control for policies that promote gender equality, such as those related to parental leave, childcare support, and flexible

[3] Alternatively, the analysis could consider additional household members beyond the heterosexual couple, such as multi-generational households. In such cases, the total household labor time contributed by all adult members, including mothers, fathers, adult children, and/or other relatives, would need to be accounted for. However, such an approach would be considerably more complex to implement. Therefore, within the scope of this study, we limit our analysis to the household labor contributions of heterosexual couples only.

working arrangements, which can help achieve greater gender equality in households. These effects are partly captured by the year fixed effects. Neither do we account explicitly for cultural attitudes towards gender roles that significantly can influence the division of household chores and responsibilities. Traditional gender roles often dictate that women are primarily responsible for household tasks, which can lead to unequal involvement in household chores and increased work-family conflict for women (Cerrato and Cifre, 2018).

4 H4

Hypothesis 4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes). That is, denoting X as a measure of support for gender equality,

$(X\text{for male respondents with daughters} - X\text{for male respondents without daughters}) >$
$(X\text{for female respondents with daughters} - X\text{for female respondents without daughters})$

4.1 Specification of the regression model

To test H4, i.e., to test the effect heterogeneity on a group (gender) level, we adjust our specification in equation 1 and include the interaction of the daughter indicator d_i with the female indicator $female_i$. Thus, we estimate the following linear model with OLS:

$$X_i = \theta_0 + \theta_1 d_i + \theta_2 d_i female_i + A_i'\delta + \epsilon_i \tag{5}$$

where the main indicator for the effect heterogeneity is the coefficient θ_2 of the interaction term. The rest of the theoretical considerations and the choice of control variable from section 1 also apply to this section.

4.2 Interpretation of results

The inclusion of the interaction effect allows for varying marginal effects between male and female parents. The men ($female = 0$) constitute the reference or the base category. The main coefficient of interest is θ_2. Thus, e.g., if $\hat{\theta_1} < 0$ and at the same time $\hat{\theta_2} = 0$, this would suggest that the effect is negative for men and that having daughters does not lead to larger effects for men than for women. If $\hat{\theta_1} > 0$ and at the same time $\hat{\theta_2} = 0$, this would suggest that the effect is positive for men and that having daughters does not lead to larger effects for men than for women. If $\hat{\theta_2}$ is statistically not significant, then we will not be able to make a statement regarding the role of daughters. On the contrary, if $\hat{\theta_2} > 0$ or $\hat{\theta_2} < 0$ (and statistically significant), this would suggest that there is effect heterogeneity. In detail, if $\hat{\theta_2} > 0$, females have a larger effect of having a daughter than males. If $\hat{\theta_2} < 0$, then there would be evidence for the contrary, and this would support the H4.

4.3 SOEP Variables to be used

4.3.1 Dependent Variable

For the choice of the dependent variable, see section 1.

4.3.2 Independent Variables

For the choice of the independent variable, see section 1.

4.4 Inclusion/exclusion of observations

For the exclusions from the analysis, see section 1.

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